Having resisted attempts to detect it directly on Earth for three-quarters of a century, dark matter remains in play. Particle physicists are confident that dark matter consists of a ghostly class of undiscovered particles that interact with matter via gravity, but otherwise interact with matter or light only weakly or not at all. If you like gambling on physics, this option is a good bet. The world's largest particle accelerators are trying to manufacture dark matter particles amid the detritus of particle collisions. And specially designed laboratories buried deep underground are trying to detect dark matter particles passively, in case they wander in from space. An underground location naturally shields the facility from known cosmic particles that might trip the detectors as dark matter impostors.

Although it all could be much ado about nothing, the idea of an elusive dark matter particle has good precedence. Neutrinos, for instance, were predicted and eventually discovered, even though they interact extremely weakly with ordinary matter. The copious flux of neutrinos from the Sun—two neutrinos for every helium nucleus fused from hydrogen in the Sun's thermonuclear core—exit the Sun unfazed by the Sun itself, travel through the vacuum of space at nearly the speed of light, then pass through Earth as though it does not exist. The tally: night and day, a hundred billion neutrinos from the Sun pass through each square inch of your body, every second, without a trace of interaction with your body's atoms. In spite of this elusivity, neutrinos are nonetheless stoppable under special circumstances. And if you can stop a particle at all, you've detected it.

Dark matter particles may reveal themselves through similarly rare interactions, or, more amazingly, they might manifest via forces other than the strong nuclear force, weak nuclear force, and electromagnetism. These three, plus gravity, complete the fab four forces of the universe, mediating all interactions between and among all known particles. So the choices are clear. Either dark matter particles must wait for us to discover and to control a new force or class of forces through which their particles interact, or else dark matter particles interact via normal forces, but with staggering weakness.

So, dark matter's effects are real. We just don't know what it is. Dark matter seems not to interact through the strong nuclear force, so it cannot make nuclei. It hasn't been found to interact through the weak nuclear force, something even elusive neutrinos do. It doesn't seem to interact with the electromagnetic force, so it doesn't make molecules and concentrate into dense balls of dark matter. Nor does it absorb or emit or